Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The wavelength of the first line in Balmer series in the hydrogen spectrum is λ. What is the wavelength of the second line.

(A) $\frac{20\lambda}{27}$

(B) $\frac{3\lambda}{16}$

(C) $\frac{5\lambda}{36}$

(D) $\frac{3\lambda}{4}$

Q.2

Which of the following series in the spectrum of the hydrogen atom lies in the visible region of the electromagnetic spectrum?

(A) Paschen series.

(B) Balmar series

(C) Lyman series

(D) Brackett series

Q.3

The distance of the closest approach of an alpha particle fired at a nucleus with momentum p is r_0. The distance of the closest approach when the alpha particle is fired at the same nucleus with momentum $2p$ will be

(A) $2r_0$

(B) $4r_0$

(C) $\frac{r_0}{2}$

(D) $\frac{r_0}{4}$

Q.4

The ratio of the wavelengths of the longest wavelength lines in the Lyman and Balmer series of hydrogen spectrum is

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(A) $\frac{3}{23}$

(B) $\frac{5}{27}$

(C) $\frac{7}{29}$

(D) $\frac{9}{31}$

Q.5

Which energy state of the triply ionized beryllium $\left(\mathrm{Be}^{+++}\right)$has the same electron orbital radius as that of the ground state of hydrogen? Given Z for beryllium $=4$

(A) $n=1$

(B) $n=2$

(C) $n=3$

(D) $n=4$

Q.6

Which energy state of doubly ionized lithium (Li $^{++}$) has the same energy as that of the ground state of hydrogen? Given Z for lithium $=3$. What is the ratio of the electron orbital radius of $\mathrm{Li}^{++}$to that of hydrogen?

(A) 1

(B) 2

(C) 3

(D) 4

Q.7

If a hydrogen atom at rest, emits a photon of wavelength λ, the recoil speed of the atom of mass m is given by

(A) $\frac{h}{m\lambda}$

(B) $\frac{mh}{\lambda}$

(C) $mh\lambda$

(D) $\frac{m}{h\lambda}$

Q.8

The energy required to excite a hydrogen atom from $n=1$ to $n=2$ energy state is $10.2\mathrm{eV}$. What is the wavelength of the radiation emitted by the atom when it goes back to its ground state?

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(A) $1020\tilde{A}\ldots$

(B) $1120\tilde{A}\ldots$

(C) $1220\tilde{A}\ldots$

(D) $1320\tilde{A}\ldots$

Q.9

The frequency of the first line in Lyman series in the hydrogen spectrum is v. What is the frequency of the corresponding line in the spectrum of doubly ionized Lithium?

(A) v

(B) $3v$

(C) $9v$

(D) $27v$

Q.10

The ionization energy of hydrogen atom is $13.6\mathrm{eV}$. Hydrogen atoms in the ground state are excited by electromagnetic radiation of energy 12.1 eV. How many spectral lines will be emitted by the hydrogen atoms?

(A) one

(B) two

(C) three

(D) four

Q.11

The ionization energy of hydrogen atom is $13.6\mathrm{eV}$ What is the ionization energy of helium atom?

(A) $3.4\mathrm{eV}$

(B) $13.6\mathrm{eV}$

(C) $54.4\mathrm{eV}$

(D) $108.8\mathrm{eV}$

Q.12

The ionization potential of the hydrogen atom is 13.6 eV. Its energy in $n = 2$ energy state is

(A) $-3.4\mathrm{eV}$

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(B) -6.8eV

(C) -13.6eV

(D) -27.2eV

Q.13

The wavelength of the second line of Balmer series is $486.4\ \text{nm}$. What is the wavelength of the first line of Lyman series?

(A) $121.6\ \text{nm}$

(B) $364.8\ \text{nm}$

(C) $729.6\ \text{nm}$

(D) none of these

Q.14

The distance of the closest approach of an alpha particle fired at a nucleus with kinetic energy K is r_0. The distance of the closest approach when the alpha particle is fired at the same nucleus with kinetic energy $2K$ will be

(A) $2r_0$

(B) $4r_0$

(C) $\frac{r_0}{2}$

(D) $\frac{r_0}{4}$

Q.15

The total energy of the electron in the first excited state of hydrogen is $-3 \cdot 4\text{eV}$. What is the kinetic energy of the electron in this state?

(A) $+1.7\text{eV}$

(B) $+3.4\text{eV}$

(C) $+6.8\text{eV}$

(D) $+13.4\text{eV}$

Q.16

In the Bohr model of the hydrogen atom, the ratio of the kinetic energy to the total energy of the electron in a quantum state n is

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(A) -1

(B) $+1$

(C) $\frac{1}{n}$

(D) $\frac{1}{n^2}$

Q.17

The energy difference between the first two levels of hydrogen atom is $10.2\mathrm{eV}$. What is the corresponding energy difference for a singly ionized helium atom?

(A) $10.2\mathrm{eV}$

(B) $20.4\mathrm{eV}$

(C) $40.8\mathrm{eV}$

(D) $81.6\mathrm{eV}$

Q.18

In Rutherford's alpha particle scattering experiment with thin gold foil, 8100 scintillations per minute are observed at an angle of 60°. The number of scintillations per minute at an angle of 120° will be

(A) 100

(B) 2025

(C) 32400

(D) 4050

Q.19

The innermost orbit of the hydrogen atom has a diameter of $1.06\tilde{A}$. What is the diameter of the tenth orbit?

(A) $5 \cdot 3\tilde{A}$

(B) $10 \cdot 6\tilde{A}$

(C) $53\tilde{A}$

(D) $106\tilde{A}$

Q.20

The total energy of the electron in the first excited state of hydrogen is $-3 \cdot 4\mathrm{eV}$. The potential energy of the electron is

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(A) -1.7eV

(B) -3.4eV

(C) -6.8eV

(D) -13.4eV

Q.21

Which energy state of doubly ionized lithium (Li^{++}) has the same energy as that of the ground state of hydrogen? Given Z for lithium $= 3$.

Q.22

The electron in the ground state of hydrogen atom is revolving in a circular orbit of radius r. If m is the mass of electron, e its charge and h the Planck's constant, the magnitude of the magnetic dipole moment of the electron is. Find x in 7x eh/88m

Q.23

The different lines in the Lyman series have their wavelengths lying between $\mathrm{n} = 2$ to $\mathrm{n} = \infty$. Find $\lambda s/300$

Q.24

The shortest wavelength in the Lyman series is $911.6\tilde{A}\ldots$. Then the longest wavelength in the Lyman series is. find $\frac{\lambda}{150}$ _____ (nearest integer)

Q.25

A proton is fired from very far away towards a nucleus with charge $Q = 120e$, where e is the electronic charge. It makes a closest approach of 10fm to the nucleus. The de Broglie wavelength (in units of fm) of the proton at its start is

$\left(\text{proton-mass } = \frac{5}{3} \times 10^{-27}\ \text{kg}, h/e = 4.2 \times 10^{-15}\ \text{J sC}^{-1}\right)$

Answers & Solutions

Q.1 Answer:

$\frac{20\lambda}{27}$

Solution:

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The first line corresponds to $n = 3$. Therefore,

$$\frac{1}{\lambda_1} = R\left(\frac{1}{4} - \frac{1}{3^2}\right)$$
$$= R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36}$$

or $\lambda_1 = \frac{36}{5R}$. We have seen above that $\lambda_2 = \frac{16}{3R}$ Hence

$$\frac{\lambda_2}{\lambda_1} = \frac{16}{3R} \times \frac{5R}{36} = \frac{20}{27}$$
$$\lambda_2 = \frac{20}{27}\lambda_1 = \frac{20}{27}\lambda$$

Q.2 Answer:

Balmar series

Solution:

The correct choice is (b).

Q.3 Answer:

$\frac{r_0}{4}$

Solution:

$K = \frac{1}{2}mv^2 = \frac{1}{2m} \times (mv)^2 = \frac{p^2}{2m}$. Therefore

$r_0 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{4mZe^2}{p^2}$

Thus $r_0 \propto \frac{1}{p^2}$. When p is doubled, r_0 becomes onefourth. Hence the correct choice is (d).

Q.4 Answer:

$\frac{5}{27}$

Solution:

The longest wavelengths in the two series are given by

$$\frac{1}{\lambda_L} = R \cdot \left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R \cdot \frac{3}{4}$$
$$\text{and} \quad \frac{1}{\lambda_B} = R \cdot \left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R \cdot \frac{5}{36}$$
$$\therefore \quad \frac{\lambda_B}{\lambda_L} = \frac{3}{4} \times \frac{36}{5} = \frac{27}{5} \text{ or } \lambda_L : \lambda_B = 5 : 27$$

Q.5 Answer:

$n = 2$

Solution:

For an atom of atomic number Z, the radius of the nth orbit is given by [see Eq.]

$r_n = \frac{Kn^2}{Z}$ **...(i)**

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where $K = \frac{\varepsilon_0 h^2}{\pi m e^2}$ is a constant. For the ground state of hydrogen $(Z = 1), n = 1$ so that $r_1 = K$

Let n be the energy state of Be^{+++} for which the orbital radius is r_1. Putting $Z = 4$ and $r_n = r_1 = K$ is Eq. (i) we get

$K = K\frac{n^2}{4}$ or $n^2 = 4$ or $n = 2$

Hence the correct choice is (b).

Q.6 Answer:

3

Solution:

Now $r_n = \frac{Kn^2}{Z}$. Therefore, for hydrogen $(n = 1$ state), we have

$r_1 = K \quad (\because Z = 1)$

and for $\mathrm{Li}^{++}(n = 3$ state) we have

$(\because Z = 3)$

$$r_3' = \frac{K(3)^2}{3} = 3K$$

$$\therefore \quad \frac{r_3'}{r_1} = \frac{3K}{K} = 3$$

Hence the correct choice is (c).

Q.7 Answer:

$\frac{h}{m\lambda}$

Solution:

Since the momentum of photon is $\frac{h}{\lambda}$, from the law of conservation of momentum, the recoil speed v of an atom of mass m is given by

$mv = \frac{h}{\lambda}$ or $v = \frac{h}{m\lambda}$

Q.8 Answer:

$1220\tilde{A}\ldots$

Solution:

Given $E_2 - E_1 = 10.2\mathrm{eV} = 10.2 \times 1.6 \times 10^{-19}$ J Therefore, frequency the emitted radiation is

$v = \frac{E_2 - E_1}{h}$

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and wavelength is

$$\lambda = \frac{c}{v} = \frac{ch}{E_2 - E_1}$$
$$= \frac{3 \times 10^8 \times 6.6 \times 10^{-34}}{10.2 \times 1.6 \times 10^{-19}}$$
$$= 1.22 \times 10^{-7}\ \mathrm{m} = 1220\tilde{A} \ldots$$

Q.9 Answer:

$9v$

Solution:

Now $v \propto Z^2$. For doubly ionized lithium $Z = 3$. Hence the correct choice is (c).

Q.10 Answer:

three

Solution:

Energy of electron in hydrogen atom in ground state $= -13 - 5\mathrm{eV}$.

When a photon of energy of $12.1\mathrm{eV}$ is radiated on atom, the energy gained by the electron is given by:

$$\mathrm{E} = (-13.6) + 12.1 = (-1.5)\mathrm{eV}$$

And we know that energy of an excited electron is given by,

$\mathrm{E} = \frac{-13.6}{\mathrm{n}^2}$ (n : **number of orbit**)

$\Rightarrow -1.5 = \frac{-13.6}{\mathrm{n}^2}$

$\Rightarrow \mathrm{n} = 3$

Q.11 Answer:

$54.4\mathrm{eV}$

Solution:

Ionization energy $E \propto Z^2$. For helium $Z = 2$. Hence E for helium $= 13.6 \times (2)^2 = 54.4\mathrm{eV}$, which is choice (c).

Q.12 Answer:

$-3.4\mathrm{eV}$

Solution:

Ionization potential = energy of the electron in the first orbit of the atom $= -13.6\mathrm{eV}$. Its value in the second orbit is

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$-\frac{13.4\text{eV}}{(2)^2} = -3.4\text{eV}$

Hence the correct choice is (a).

Q.13 Answer:

121.6 nm

Solution:

Wavelengths in Balmer series for hydrogen are given by

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$$
$$= R\left(\frac{1}{4} - \frac{1}{n^2}\right); n = 3, 4, 5 \ldots$$

The second line in Balmer series corresponds to $n = 4$. Hence

$\frac{1}{\lambda_2} = R\left(\frac{1}{4} - \frac{1}{16}\right) = \frac{3R}{16}$ or $\lambda_2 = \frac{16}{3R}$

The wavelength of the first line $(n = 2)$ in Lyman series is

$$\frac{1}{\lambda_1} = R\left(1 - \frac{1}{2^2}\right) = R\left(1 - \frac{1}{4}\right)$$
$$= \frac{3R}{4} \text{ or } \lambda_1 = \frac{4}{3R}$$
$$\therefore \quad \frac{\lambda_1}{\lambda_2} = \frac{4}{3R} \times \frac{3R}{16} = \frac{1}{4}$$
$$\text{or } \lambda_1 = \frac{\lambda_2}{4} = \frac{486.4}{4} = 121.6 \text{ nm}$$

Hence the correct choice is (a).

Q.14 Answer:

$\frac{r_0}{2}$

Solution:

The distance of the closest approach is given by

$r_0 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{2Ze^2}{K}$

where $K = \frac{1}{2}mv^2$. Thus $r_0 \propto \frac{1}{K}$. When K is doubled, r_0 becomes half. Hence the correct choice is (c).

Q.15 Answer:

$+3.4\text{eV}$

Solution:

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The kinetic and potential energies of an electron in the nth excited state are given by

$$\text{KE} = \frac{1}{8\pi\varepsilon_0}\cdot\frac{e^2}{r_n} \quad ...(\text{i})$$

$$\text{and PE} = -\frac{1}{4\pi\varepsilon_0}\cdot\frac{e^2}{r_n} \quad ...(\text{ii})$$

Adding (i) and (ii) we get the total energy E which is

$E = \text{KE} + \text{PE} = -\frac{1}{8\pi\varepsilon_0}\cdot\frac{e^2}{r_n}$ **...(iii)**

Notice from (i) and (iii) that $E = -\text{KE}$. Given $E = -3.4\text{eV}$. Hence $\text{KE} = -E = -(-3.4) = +3.4\text{eV}$ Thus the correct choice is (b).

Q.16 Answer:

-1

Solution:

The total energy of an electron bound to an atom is negative and is the sum of its P.E. and K.E. The magnitude of the P.E. is twice that of the K.E. (as per the Bohr model) but since the P.E. is -ve, we have Total Energy $= (-2K) + (K) = -K$ ($K =$ Kinetic energy). Hence the ratio of total energy and kinetic energy $= -1 : 1 = -1$

Q.17 Answer:

40.8eV

Solution:

Energy difference $\Delta E \propto Z^2$. For a singly ionized helium atom $Z = 2$.

Hence $\Delta E = 10.2 \times (2)^2 = 40.8\text{eV}$

Q.18 Answer:

100

Solution:

The number of scintillations per unit time is $n \propto \cot^4\left(\frac{\theta}{2}\right)$. Hence

$$\frac{n_2}{n_1} = \frac{\cot^4(120^\circ/2)}{\cot^4(60^\circ/2)} = \frac{\cot^4(60^\circ)}{\cot^4(30^\circ)}$$

$$= \left(\frac{1/\sqrt{3}}{\sqrt{3}}\right)^4 = \frac{1}{81}$$

Hence $n_2 = \frac{1}{81} \times 8100 = 100$, which is choice (a).

Q.19 Answer:

$106\tilde{A}$

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Solution:

The diameter of nth orbit is proportional to n^2, i.e. $d_n \propto n^2$. Given $d_1 = 1.06\tilde{A}\ldots$. Therefore $d_{10} = (10)^2 d_1 = 106\tilde{A}\ldots$. Hence the correct choice is (d).

Q.20 Answer:

-6.8eV

Solution:

From (ii) and (iii) we find that $E = \frac{\text{PE}}{2}$ or $\text{PE} = 2E = 2 \times -3.4 = -6.8\text{eV}$.
Hence the correct choice is (c).

Q.21 Answer:

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Solution:

$E_{\text{n}} = \frac{CZ^2}{n^2}$. For $n = 1$ state of hydrogen, we have $E_1 = C$ and for the nth state of Li^{++}, we have $E_n = \frac{C(3)^2}{n^2} = \frac{9C}{n^2}$. For $E_1 = E_n$, we require $C = \frac{9C}{n^2}$ which gives $n = 3$.

Q.22 Answer:

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Solution:

From Bohr's quantization condition,

$mvr = \frac{nh}{2\pi}$ (1)

where v is the speed of the electron, which from (1) is given by

$v = \frac{nh}{2\pi mr}$

For ground state, $n = 1$. Hence

$v = \frac{h}{2\pi mr}$ (2)

Current around the orbit is (here T = time period)

Current around the orbit is (here T = time period)

$$I = \frac{e}{T}$$

$$\text{But } T = \frac{2\pi r}{v}$$

$$\therefore \quad I = \frac{ev}{2\pi r} \quad (3)$$

Using (2) in (3), we get

$I = \frac{eh}{4\pi^2 mr^2}$

Magnetic dipole moment $= IA = I \times \pi r^2$

$= \frac{eh}{4\pi^2 mr^2} \times \pi r^2$

$= \frac{eh}{4\pi m}$

Q.23 Answer:

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Solution:

For Lyman series

$\frac{1}{\lambda}=R\left(\frac{1}{1^2}-\frac{1}{n^2}\right) ; n=2,3,4 \ldots$

The extreme wavelengths correspond to $n=2$ **and** $n=\infty$

Therefore, the wavelength limits are

$\lambda_s=\frac{1}{R}$ **and** $\lambda_l=\frac{4}{3R}$**. Since** $\frac{1}{R}=\frac{1}{1.1\times10^7}\ \mathrm{m} \cong 0.9\times10^{-7}\ \mathrm{m}=900\tilde{A} \ldots$**, we have**

$\lambda_l=\frac{4}{3}\times900=1200\tilde{A} \ldots$ and $\lambda_s=900\tilde{A} \ldots$

Q.24 Answer:

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Solution:

For Lyman series, we have $\frac{\lambda_l}{\lambda_s}=\frac{4}{3}$**. Hence**

$\lambda_l=\frac{4}{3}\times911.6=1215\tilde{A} \ldots$

Q.25 Answer:

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Solution:

$$\frac{p^2}{2m}=\frac{KZe^2}{r} \Rightarrow p=\sqrt{\frac{KZe^2\cdot2m}{r}}$$

$$\Rightarrow \lambda=\frac{h}{p}=\frac{h/e}{\sqrt{2K\frac{Zm}{r}}}=\frac{4.2\times10^{-15}}{\sqrt{\frac{2\times9\times10^9\times120\times\frac{5}{3}\times10^{-27}}{10\times10^{-15}}}}=7\mathrm{fm}.$$

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